

Related Work Table and Research Gap

Md. Rafiqul Islam (2026-2-74-025)
Ahanaf Shahadat Taseen (2026-2-74-033)
Reasad Mahmood (2026-2-74-012)

Department of Computer Science and Engineering, East West University

Summer 2026

The reviewed studies serve two different roles. Attention-based malware-image papers provide architectural evidence for channel, spatial and self-attention, whereas SOMLAP papers provide dataset-specific performance context. Results are reported with their original protocols and are not treated as paired comparisons with the current duplicate-group held-out set.

Authors	Year	Title	Dataset	Application area	Method/model	Attention type	Metrics	Strengths	Limitations	Research gap	Relation to our work
Wang et al.	2022	MSAAM: A Multiscale Adaptive Attention Module for IoT Malware Detection and Family Classification	MalImg; BIG 2015	Malware detection and family classification	CliqueNet with MSAAM	Hybrid channel and spatial	Detection 99.8%; family accuracy 99.2% and 98.2%	Adaptive multiscale weighting; imbalance experiments	Large image model; small binary subset	No test on PE-header feature grids	Supplies the channel-spatial design rationale
Kattamuri et al.	2023	Swarm Optimization and Machine Learning Applied to PE Malware Detection towards Cyber Threat Intelligence	SOMLAP	Binary PE malware detection	ACO, CSO and GWO feature selection with classical ML	None	Best accuracy 99.37% using ACO and 12 features	Introduces SOMLAP and reports feature-selection experiments	Published split cannot be reproduced from the supplied file	No CNN or attention model	Dataset source and contextual benchmark

Authors	Year	Title	Dataset	Application area	Method/model	Attention type	Metrics	Strengths	Limitations	Research gap	Relation to our work
Ravi and Alazab	2023	Attention-based convolutional neural network deep learning approach for robust malware classification	MalImg-style images, 25 families	Malware family classification	CNN with multi-head attention	Multi-head self-attention	About 99% accuracy	Direct comparison between attention CNN and plain variants	Byte images and a different label space	No SOMLAP or matched dropout control	Supports a controlled attention experiment
Huang et al.	2023	A Malicious Code Detection Method Based on Stacked Depthwise Separable Convolutions and Attention Mechanism	MalImg; Blended+	Malware detection and family classification	CoAtNet with depthwise separable convolutions	Self-attention with convolution	99.33% and 96.60% accuracy	Tests an easy and a harder image set	Lower performance on Blended+; no PE-header data	Synthetic feature-grid behavior remains unknown	Provides a convolution-attention comparison point
Fan et al.	2025	GCSA-ResNet: a deep neural network architecture for malware detection	MalImg; BIG 2015	Malware family classification	ResNet with global channel-spatial attention	Channel and spatial	99.08% and 98.58% accuracy	Reports false-positive behavior on two datasets	Image-only family classification	No evaluation on static header tables	Closest recent channel-spatial architecture
Liu et al.	2026	Research on image malware classification based on AlexNet transfer learning and MHSA	MaleVis	Malware family classification	AlexNet transfer learning with MHSA	Multi-head self-attention	87.3% accuracy	Includes Grad-CAM examples	Short conference paper; moderate result	No SOMLAP test or layout sensitivity study	Shows a heavier attention option

Authors	Year	Title	Dataset	Application area	Method/model	Attention type	Metrics	Strengths	Limitations	Research gap	Relation to our work
Alwateer et al.	2026	A hierarchical deep learning framework with doubly regularized loss for malware detection and family categorization	SOMLAP and other PE datasets	Binary detection and family categorization	Hierarchical deep network with DRBCE loss	None stated	SOMLAP accuracy 99.11%; weighted F1 98.85%	Several metrics and cross-dataset tests	No shared predictions or duplicate-group split	No attention and limited field-level explanation	Strongest verified SOMLAP context; not a paired comparison

Table 1. Reviewed CNN, attention and SOMLAP studies published from 2022 to 2026.

Research gap

Most recent malware-attention models operate on byte-derived images and predict malware families. SOMLAP instead contains numeric PE-header fields and has been evaluated mainly with feature selection or deep tabular networks. The literature does not establish whether a lightweight channel-spatial module improves a matched CNN when those 108 fields are placed on a fixed grid.

The experimental gap has three parts. First, attention must be separated from dropout and other training choices through matched controls. Second, exact duplicate vectors must remain within one partition because the supplied file contains substantial duplication. Third, spatial explanations must be mapped to named feature slots without presenting the grid as a physical executable image. The project addresses these points through a factorial attention-by-dropout comparison, grouped five-fold evaluation and both Grad-CAM and LIME.

Dataset-count correction

The SOMLAP source article reports 51,409 records, including 19,809 malware samples. The supplied CSV contains 51,408 records and 19,808 malware samples. All descriptive statistics, partitions and model results in this project use the file actually analyzed. The 2026 hierarchical study reports 99.11% accuracy and 98.85% weighted F1 on SOMLAP; its 99.89% value belongs to a different evaluation and is not used as the SOMLAP benchmark.

References

- [1] C. Wang, Z. Zhao, F. Wang, and Q. Li, 'MSAAM: A Multiscale Adaptive Attention Module for IoT Malware Detection and Family Classification,' Security and Communication Networks, 2022, doi: 10.1155/2022/2206917.
- [2] S. J. Kattamuri, R. K. V. Penmatsa, S. Chakravarty, and V. S. P. Madabathula, 'Swarm Optimization and Machine Learning Applied to PE Malware Detection towards Cyber Threat Intelligence,' Electronics, vol. 12, no. 2, art. 342, 2023, doi: 10.3390/electronics12020342.
- [3] V. Ravi and M. Alazab, 'Attention-based convolutional neural network deep learning approach for robust malware classification,' Computational Intelligence, 2023, doi: 10.1111/coin.12551.

- [4] H. Huang, R. Du, Z. Wang, et al., 'A Malicious Code Detection Method Based on Stacked Depthwise Separable Convolutions and Attention Mechanism,' Sensors, vol. 23, no. 16, art. 7084, 2023, doi: 10.3390/s23167084.
- [5] Y. Fan, K. Zhang, B. Zheng, et al., 'GCSA-ResNet: a deep neural network architecture for Malware detection,' Scientific Reports, 2025, doi: 10.1038/s41598-025-10561-6.
- [6] J. Liu, G. Zhou, Y. He, and X. Hou, 'Research on image malware classification method based on AlexNet transfer learning and MHSA attention mechanism,' Proc. SPIE 14122, 141220X, 2026, doi: 10.1117/12.3105850.
- [7] M. Alwateer et al., 'A hierarchical deep learning framework with doubly regularized loss for robust malware detection and family categorization,' Scientific Reports, 2026, doi: 10.1038/s41598-025-31444-w.